

|         |                                                                                              |    |     |
|---------|----------------------------------------------------------------------------------------------|----|-----|
| 7 (a)   | photoelectric effect .....                                                                   | B1 | [1] |
| (b) (i) | reasonable line extrapolated .....                                                           | C1 |     |
|         | $6.8 \times 10^{14}$ Hz (allow $\pm 0.4 \times 10^{14}$ Hz) .....                            | A1 |     |
| (ii)    | attempt at finding gradient .....                                                            | M1 |     |
|         | working shown to give $6.6 \times 10^{-34}$ J s Hz (allow $\pm 0.4 \times 10^{14}$ Hz) ..... | A1 | [4] |
| (c)     | line: same gradient .....                                                                    | B1 |     |
|         | to the left of the line drawn by candidate .....                                             | B1 | [2] |
| (d)     | maximum corresponds to electron emitted from surface .....                                   | B1 |     |
|         | other electrons require energy to be brought to the surface .....                            | B1 | [2] |